Venkata Sai

Software Engineer

venkatasai.jobs@gmail.com | +1(512)-848-0295 | Austin, TX

Professional Experience

Senior Software Engineer *Jul 2025 – Present*

Persistent Systems | AI-Powered Media Personalization Platform

- Led development of **Deep Learning** models using **TensorFlow** and **PyTorch** for Prime Video search systems, improving content relevance by 18% in A/B tests.
- Architected and implemented **Generative AI** solutions with **Hugging Face** and **OpenAI API** to personalize content summaries, boosting user engagement by 15%.
- Designed and executed **Reinforcement Learning** algorithms in **Python** on **AWS SageMaker** for dynamic recommendation feeds, increasing conversion rates by 12%.
- Optimized critical **machine learning** pipelines using **Kubernetes** and **Airflow** on **AWS EKS**, processing 50TB daily, reducing inference latency by 30%.
- Collaborated cross-functionally to define **AI solution** roadmaps for Prime Video's discovery platforms, integrating **data analyses** to achieve 25% feature adoption.
- Implemented robust **A/B testing** frameworks for new **AI models** using **AWS S3** and **Spark**, ensuring data-driven decisions and 10% uplift in key metrics.

Software Engineer *Jul 2022 – Jun 2025*

Nagarro | Enterprise AI & Data Platform

- Developed scalable **Python** backend services for a FinTech data platform using **Flask** and **PostgreSQL**, handling 10,000 requests/second with 99.9% uptime.
- Engineered **machine learning** prediction systems for fraud detection leveraging **Scikit-learn** and **AWS Lambda**, reducing false positives by 22% monthly.
- Integrated **OpenAI API** with custom **Python** services to generate dynamic content for marketing campaigns, increasing click-through rates by 18% post-November 2022.
- Built **ETL pipelines** using **Apache Kafka** and **AWS Kinesis** for real-time data ingestion, processing 100M events daily, improving data freshness by 50%.
- Designed and deployed a search ranking algorithm using **Elasticsearch** and **MLflow**, enhancing query relevance by 15% and user satisfaction scores by 10%.
- Implemented **LangGraph**-based agentic AI workflows with **Python** and **Redis** for automated customer support, reducing response times by 35% after January 2024.
- Optimized database queries and **data models** in **PostgreSQL** for a high-traffic e-commerce application, decreasing average page load time by 20%.
- Utilized **Docker** and **Kubernetes** to containerize and orchestrate microservices, achieving a 30% improvement in deployment efficiency and resource utilization.

Software Engineer *Jul 2020 – Aug 2022*

Endava | Digital Content Delivery System

- Developed and maintained **Java** microservices for a digital content delivery platform using **Spring Boot**, supporting 500,000 concurrent users reliably.
- Implemented robust **RESTful APIs** for video streaming metadata, ensuring data consistency and low latency responses for over 2 million daily requests.
- Designed and optimized **SQL** database schemas with **MySQL** for content catalog management, improving query performance and data retrieval speed by 40%.
- Automated **CI/CD pipelines** using **Jenkins** and **Git** for faster software releases, reducing deployment time from hours to minutes by 70%.
- Contributed to **system design** and architecture reviews, focusing on **scalability** and **reliability** for critical components, preventing 95% of potential outages.
- Utilized **Jira** for agile project management and issue tracking, collaborating with cross-functional teams to deliver features on schedule 90% of the time.

Skills

Languages: Python, Java, SQL
Frameworks & Tools: TensorFlow, PyTorch, Hugging Face, Scikit-learn, Flask, Apache Kafka, AWS Kinesis, MLflow, LangGraph, Redis, Docker, Kubernetes, Spring Boot, Jenkins, Git, Jira, Spark, Airflow
Cloud & Infra: AWS SageMaker, AWS EKS, AWS Lambda, AWS S3
Databases & Search: PostgreSQL, Elasticsearch, MySQL
Concepts: System Design, Scalability, Reliability, A/B Testing, ETL Pipelines, Data Modeling, Microservices, CI/CD, API Design, Optimization, Data Analyses, Cross-functional Collaboration
AI Tools: Deep Learning, Generative AI, Reinforcement Learning, Machine Learning Models, OpenAI API

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India